

Outcomes of Choroidal Melanoma 25-Gauge Vitrectomy Needle Biopsy and Iodine 125 Brachytherapy Performed by Directly Supervised Fellows versus Attending Faculty

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Purpose: To evaluate the outcomes of choroidal melanoma 25-gauge vitrectomy needle biopsy and iodine 125 (¹²⁵I) brachytherapy performed by supervised fellows compared with experienced faculty.

Design: Retrospective, observational case series.

Participants: Sixty-three consecutive eyes of 63 patients with treatment-naïve medium choroidal melanomas without extraocular extension.

Methods: Sixty-three consecutive eyes of 63 patients undergoing simultaneous ¹²⁵I brachytherapy and 25-gauge vitrectomy melanoma biopsy for choroidal melanoma. One attending surgeon performed 31 surgeries and 2 fellows performed 32 surgeries. All cases treated by fellows were under direct supervision of the attending physician.

Main Outcome Measures: Percentage of adequate biopsy sample, tumor regression, and postoperative complications.

Results: There were no baseline differences between the groups. The biopsy yield for gene expression profile and the melanoma regression rate were equal in both groups (63/63 [100%]; $P = 1.0$). Complications occurred in 1 of 31 faculty surgery patients (transient vitreous hemorrhage) and in 0 of 32 fellow surgery patients ($P = 1.0$).

Conclusions: With proper supervision, fellows can achieve equally high biopsy yield and melanoma regression rate with few complications compared with experienced faculty. *Ophthalmology Retina* 2017;1:555-558 © 2017 by the American Academy of Ophthalmology

Choroidal melanoma 25-gauge vitrectomy needle biopsy with iodine 125 (¹²⁵I) brachytherapy is an innovative biopsy technique used in the treatment of ocular melanoma. A 25-gauge pars plana vitrectomy (PPV) is performed with needle biopsy to increase tissue yield and prevent vision loss through controlled hemostasis. A vitreoretinal fellowship requires adequate experience, a sizable number of surgical procedures performed, and proper teaching to attain the experience necessary for excellence as an attending surgeon. The safety and efficacy of this particular procedure performed by vitreoretinal fellows are 2 concerns that arise when compared with fellowship-trained attending physicians.

Fellow versus attending surgical outcomes have been reported in the literature in several areas of ophthalmology, including 25-gauge vitrectomy,¹⁻³ penetrating keratoplasty,⁴ blepharoptosis repair,⁵ and cataract surgery.⁶ However, this is the first study to evaluate ocular choroidal melanoma surgical outcomes regarding 25-gauge vitrectomy with needle biopsy and ¹²⁵I brachytherapy. Mason et al¹ found that with proper supervision, fellows can achieve equally low complication and reoperation rates with vitreoretinal surgeries compared with those of attending physicians with regard to PPV for macular hole, macular

pucker, retinal detachment, diabetic vitreous hemorrhage, and diabetic traction retinal detachment. Our purpose was to compare complication rate, biopsy yield, and melanoma regression rate after PPV with needle biopsy and brachytherapy between supervised vitreoretinal fellows and their attending staff.

Methods

Approval from the University of Alabama at Birmingham Institutional Review Board and informed consent were obtained before collection of patient data. The study complied with the Health Insurance Portability and Accountability Act and adhered to the tenets of the Declaration of Helsinki.

In this retrospective cohort study, we analyzed 63 consecutive eyes of 63 patients who underwent simultaneous brachytherapy and 25-gauge vitrectomy needle biopsy for melanoma at the University of Alabama at Birmingham and Retina Consultants of Alabama from July 2012 through July 2015. Two fellows (J.N.C. and D.M.R.) performed 32 procedures, whereas 1 attending (J.O.M.) performed the remaining 31 procedures. All surgeries by fellows were performed under the direct supervision and teaching of the attending physician involved in the study. Supervised procedures were cases in which complete guidance was provided, with the absence of direct surgical intervention by the attending. All